

Assignment 3

SPSS For Research

Instructor: Md. Ahsanul Islam

Deadline: 29 August 2026

Submission: Save your assignment as a word document named Assignment 3_YOURNAME.docx, write your name (as per certificate) at the top, and submit the word file.

Problem Statement:

The dataset assignment3.sav contains records for 400 mothers who recently gave birth, collected in a maternal and child health survey. For each mother the file records place of residence, household wealth, age, the number of antenatal care (ANC) visits made during pregnancy, and whether the baby was born with low birth weight (below 2500 g). Use this one file for every question below.

Variable	Label	Values / Units	Measure
mother_id	Mother identification number	Integer ID	Nominal
low_birth_weight	Baby born with low birth weight	0 = Normal (≥ 2500 g); 1 = Low (< 2500 g)	Nominal
mother_age	Mother's age	Years	Scale
anc_visits	Number of antenatal care (ANC) visits during pregnancy	Count	Scale
wealth_index	Household wealth index	1 = Poor; 2 = Middle; 3 = Rich	Ordinal
residence	Place of residence	0 = Rural; 1 = Urban	Nominal

Question 1

Produce descriptive statistics for the sample and complete **Table 1**. For the categorical variables report the frequency and percentage of each category; for the two continuous variables report the mean (standard deviation), median, minimum and maximum.

Table 1. Descriptive characteristics of the sample (N = ____).

Variable	Level / statistic	Frequency, n (%)	Mean (SD)	Median	Minimum	Maximum
Low birth weight	Normal (≥ 2500 g)		—	—	—	—
	Low (< 2500 g)		—	—	—	—

Residence	Rural		—	—	—	—
	Urban		—	—	—	—
Wealth index	Poor		—	—	—	—
	Middle		—	—	—	—
	Rich		—	—	—	—
Mother's age (years)		—				
ANC visits		—				

Cells marked — are not applicable to that variable and should be left blank.

Question 2

Test whether the mean number of ANC visits differs between mothers living in rural and urban areas. State the null and alternative hypotheses, then run Independent-Samples T Test with anc_visits as the Test Variable and residence as the Grouping Variable (Define Groups: 0 and 1). Use Levene's test to decide which row to read, complete Table 2, and write a one-sentence conclusion in APA style.

Table 2a. Group statistics for ANC visits by residence.

Residence group	n	Mean	Std. Deviation	Std. Error Mean
Rural				
Urban				

Table 2b. Independent-samples t-test result.

Independent-samples t-test statistic	Value
Levene's Test for Equality of Variances - F	
Levene's Test P-value	
t	
df	
P-value	
Mean difference (Urban - Rural)	
Std. error difference	
95% Confidence Interval	

Question 3

Fit a binary logistic regression predicting low birth weight from mother's age, number of ANC visits, wealth index and residence.

Complete Table 3 from the SPSS's output and also explain the relationships between the independent variables and low birth weight.

Table 3. Binary logistic regression predicting low birth weight (1 = Low).

Predictor	Estimate (B)	SE	Odds ratio Exp(B)	95% CI for OR	p- value
Mother's age					
ANC visits					
Wealth index: Poor	— reference —				
Wealth index: Middle					
Wealth index: Rich					
Residence: Rural	— reference —				
Residence: Urban					
Constant					

Reference categories are marked. Also report the overall model fit below the table:

Omnibus test of model coefficients: $\chi^2(\text{---}) = \text{---}$, $p = \text{---}$. $-2 \text{ Log-likelihood} = \text{---}$.
Nagelkerke $R^2 = \text{---}$.

Hosmer–Lemeshow test: $\chi^2(\text{---}) = \text{---}$, $p = \text{---}$. Overall percentage correctly classified =
_____ %.